$\frac{c}{3}(64-1) = 1$

$c \quad \frac{63}{3}(21) = 1$

$c = \frac{1}{21}$

$$\int_2^3 \frac{1}{21} \cdot x^2 \, dx$$

$$\Rightarrow \frac{1}{21(3)} (27-8)$$

$$\frac{19}{21(3)}$$

$$0.3015$$

CDF

$$P(X) \begin{cases} Cx^2 & 1 \le x \le 4 \\ 0 & x < 1 \text{ or } x > 4 \end{cases}$$

$$F(x) = P(X \le x),$$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$= \int_{-\infty}^{1} 0 \, dt + \int_1^x \frac{1}{21} \cdot x^2 \cdot dx$$

$$= \frac{1}{63} \left[\frac{x^3 - 1}{3} \right]$$

$$\text{CDF} \Rightarrow \begin{cases} 0 & x < 1 \\ \frac{x^3 - 1}{63} & 1 \le x \le 4 \\ 1 & x > 4 \end{cases}$$